

Synchronizing QC bias ramps with Presto acquisition

PRESTO-8 + AGILENT 33220A | 15 SEPTEMBER 2026

Recommendation

Test a small change to the Presto startup sequence first. Keep the existing AWG gated sawtooth, defer the synchronization in `apply_settings(auto_sync=False)`, and let `get_pixels(..., auto_sync=True)` issue the start event after acquisition has been armed. The installed Presto 2.17.1 source supports this design. It is a credible solution, but its exact hardware timing still needs measurement.

The present code starts the gate before it requests pixels. Host scheduling and instrument command latency lie between those operations. The proposal moves that variable interval before the shared start; it does not try to estimate and subtract it afterward.

Assessment	What can be concluded
High confidence	The current call sequence contains an uncontrolled timing gap. Fixed sleeps or a fixed 50 ms correction cannot make it deterministic.
Strong candidate	A shared internal SYSREF event can plausibly release generation and acquisition together. Public API documentation and installed source support testing this.
Still unverified	Gate behavior before release, the first returned pixel’s timing, startup jitter, and the actual bias waveform at the device.

A synchronization fix also needs correct voltage bookkeeping. The wrapper drops 25 ms, the Presto library drops one initial pixel internally, and folding starts its displayed time at zero. Long records additionally require clock-drift control or measured ramp phase.

Scope and evidence

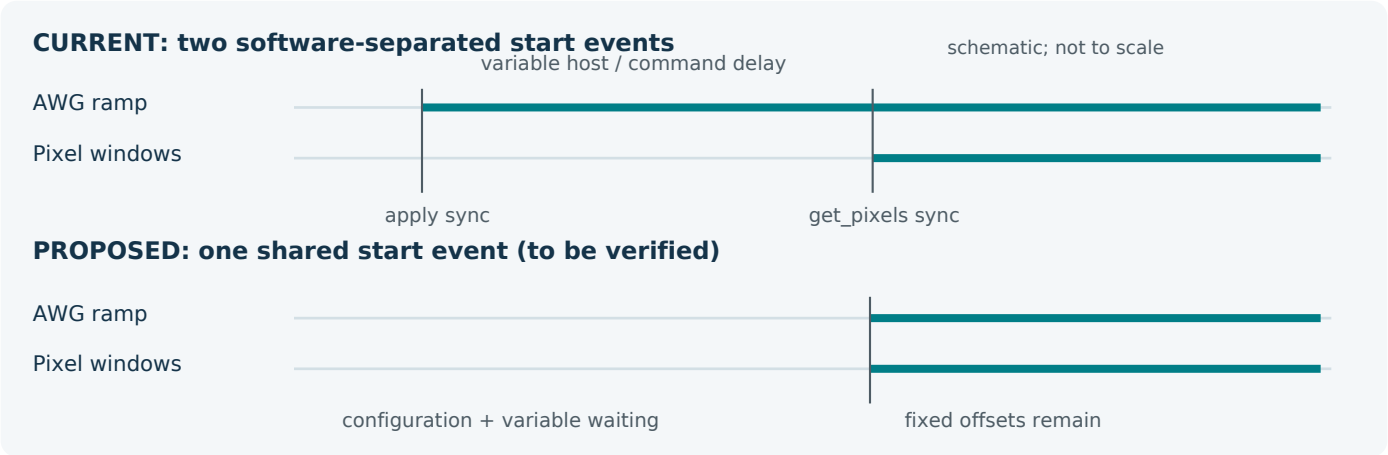
This report audits the repository and official documentation; it does not certify an instrument measurement. No acquisition, instrument connection, firmware update, or measurement-code edit was performed. The local API is 2.17.1; connected instrument firmware and AWG options were not queried. API and hardware evidence are separated from engineering proposals throughout.

Reading order: diagnosis (p.2), minimal fix (p.3), AWG and wiring (p.4), voltage model (p.5), drift (p.6), alternatives (p.7), validation (p.8), implementation boundaries (p.9), sources (p.10).

Where the timing gap enters

The current QC path is **QCTrace.run** → **GateBiasMeasurement._run_gated_ramp** → **TimeStream.run**. StdDevSweep shares this ramp-acquisition machinery, so the same issue applies there. The archived qct_measurement.py uses the same gated-ramp convention.

Stage	Observed code behavior
AWG preparation	_gate_bias.py:278 calls bias.sawtooth(gated=True); the AWG is armed before TimeStream connects and configures Presto.
Trigger setup	timestream.py:366 stages the selected digital outputs with a 30 ms width.
Early start	timestream.py:368 calls apply_settings() with its default synchronization enabled.
Later acquisition	timestream.py:378 calls get_pixels(). An optional on_acquire callback can add more work between the two calls.
Analysis origin	TimeStream saves the untrimmed return data, trims its in-memory arrays, and folding.py:104 creates a new zero-based time axis.



The source shows additional command handling after the early start, including a one-window hardware wait and RF unblanking. Its timing is not a calibrated offset. The observed O(50 ms) variability is consistent with this architecture; this audit cannot identify how much comes from each host, network, or server operation.

The wrapper’s “external_trigger” is a **Presto output** controlling another instrument. It does not mean the Lockin receiver is waiting for an external TTL acquisition trigger. The documented get_pixels return contains quadratures and frequencies, without a per-pixel trigger timestamp. [1]

The smallest credible fix

Proposed experiment — not applied to the repository

```
# Finish tuning and input/output-group configuration first.
# Establish a low gate; prepare and arm the AWG deliberately.
lck.set_trigger_out(trigger_states, width=0.03)
lck.apply_settings(auto_sync=False)
# No intervening operation may issue a synchronization event.
pixel_dict = lck.get_pixels(pixel_counts, auto_sync=True)
```

SYSREF is Presto's internal synchronization event. In the proposal, variable Python or network delay before `get_pixels` changes when the experiment begins, while the hardware event establishes the relative start of the ramp and acquisition.

Why this is supported

The public API describes `apply_settings(auto_sync=False)` as deferring generation synchronization; `get_pixels(auto_sync=True)` internally synchronizes acquisition. It does not explicitly specify the combined sequence's first-gate-edge to first-returned-pixel delay. [1]

The installed source is more informative: `lockin.py:283–307` stages trigger settings and conditionally emits `hardware.sync()`; `:877–882` arms pixel acquisition and then emits `hardware.sync()`. Both paths refer to SYSREF. Deferred RF configuration also receives `do_sync=False` (`:222–238`). This supports a common release, but Python source is not a substitute for FPGA timing verification.

Conditions for a valid test

Start from a verified low gate. A fresh Python object does not establish the electrical idle state. Complete configuration while the AWG cannot respond to unwanted edges, then arm it with the line low. Check the full preparation sequence for unintended syncs.

Retain real pending IF settings. In 2.17.1, `set_trigger_out` does not itself mark IF configuration dirty. Initial `TimeStream` setup does; a later trigger-only update may need a public setter such as `set_df(lck.get_df())` before `apply_settings`. Do not rely on a silently unsent update.

Measure startup settling again. RF generation may now start alongside acquisition. The old leading-data discard may still be useful, but its required duration and phase effect must be measured. Keep the normal immediate cleanup behavior; do not change every `apply_settings` call to defer synchronization.

Decision: accept this method only if a recorded timing witness stays at the same pixel position when deliberate host delays are inserted before `get_pixels`. See p.8.

AWG behavior and the physical link

What the manufacturer documents

The 33220A's gated burst runs while the gate is active; after deassertion it finishes the cycle and holds its start-phase voltage. Trigger-source selection is ignored in this mode. Its front Sync output follows the gate/burst envelope, not every ramp cycle. Continuous and infinite-count burst operation provide cycle timing there. [3] (pp.68-69, 107-108).

Ramp phase zero is the rising offset crossing; phase units are configurable. Trigger specifications list latency below 500 ns and 6 ns RMS jitter, with TTL pulses longer than 100 ns into a DC-coupled input above 10 k Ω . These are instrument specifications, not end-to-end alignment guarantees. [3] (pp.112, 348).

What that means for this measurement

Keep the existing gated mode for the first test. Switching to an edge-triggered burst while retaining the early Presto start would retain the software timing gap. A bus trigger, including *TRG, issued just before get_pixels would also leave two software-controlled starts.

The wrapper chooses 100% ramp symmetry, phase 180°, and a default offset of half the peak-to-peak amplitude. For an ideal rising sawtooth, 180° lands at rollover. Calibrate the actual first ramp and held level instead of assigning an endpoint from that discontinuity. A future implementation should explicitly select degree units before programming phase and read back the state it depends on.

Wiring to verify at the receiving connector

```
Presto selected DIGITAL OUT ---> 33220A rear TRIG / BURST GATE
33220A analog OUTPUT      ---> existing device bias line
Scope: observe gate and actual analog ramp together
```

Presto's datasheet lists a 3.3 V digital output with 50 Ω output impedance. Establish the loaded voltage, overshoot, cable termination, and accepted AWG input levels at the actual connector; do not infer compatibility from "TTL" alone or assume whether 3.3 V denotes open-circuit or terminated operation. [2] (p.4).

The 30 ms gate width is the existing configuration, not a verified continuous-level command. At 50 kHz, windows are 20 μ s apart. The repository infers that overlapping triggers keep the line high and explicitly says this has not been scope-verified. Look for narrow low glitches and verify rearming and gate-off behavior. At much lower df, the same width can produce separated gates.

Assigning voltage to each pixel

Repeatability and absolute voltage are separate requirements. A fixed delay can be calibrated. A variable delay must be eliminated or measured per record. Even perfect timing cannot turn an averaged pixel into an instantaneous measurement.

Use a calibrated time origin

Let f_s be the tuned Lockin pixel rate, f_r the measured ramp frequency, and $D = \text{round}(\text{discard_start_ms} \times 10^{-3} \times f_s)$. Let t_c be the measured center time of returned raw pixel 0 relative to the gate edge. For retained pixel j :

$$t_{\text{center}}(j) = t_c + (D + j) / f_s$$

The installed library requests and removes one initial pixel when no FIR is used (lockin.py:794–820, :905). This does **not** prove $t_c = 1.5/f_s$: the discarded pixel may represent pipeline or partial-window behavior. Measure t_c or obtain its definition from Intermod. Folded data must preserve D and t_c rather than treating retained sample zero as the gate edge.

At the lab defaults, $D = 1,250$ and $D/f_s = 25$ ms. A 500 Hz ramp has a 2 ms period, so trimming alone moves the origin by **12.5 periods: half a cycle modulo the ramp**. Hardware synchronization does not remove that phase shift.

Ideal ramp model, away from reset

$$\begin{aligned} u(j) &= \text{frac}[f_r * (t_{\text{center}}(j) - \tau) + u_{\text{start}}] \\ V_{\text{source}}(j) &= V_{\text{offset}} + V_{\text{pp}} * (u(j) - 0.5) \end{aligned}$$

Here frac keeps the fractional part, τ is the calibrated gate-to-ramp delay, and u_{start} is the measured initial phase. This formula assumes a 100% rising sawtooth. Other symmetry settings need a piecewise rise/fall model. Device voltage additionally depends on bias-line gain, offset, filtering, and propagation delay; a fixed delay is inadequate if the line distorts the ramp.

Each unsummed pixel integrates over roughly $1/f_s$. On a locally linear ramp the voltage traversed in that interval is $V_{\text{pp}} f_r / f_s$. For the illustrative 2 Vpp, 500 Hz ramp at 50 kHz, this is **20 mV per pixel**. A midpoint voltage is a useful label, not an exact description of all voltages contributing to the result. Exclude reset-spanning and unsettled windows or model their response.

Away from reset, a timing uncertainty δt produces approximately $\delta V = V_{\text{pp}} f_r \delta t$. Thus a 1 mV timing-error budget in this example requires about 1 μs timing knowledge. This is distinct from the 20 μs integration width.

Prevent phase drift during the record

One start event does not lock two clocks

A relative clock error ϵ accumulates a time slip of approximately ϵT over a record of duration T . As an illustrative calculation, 20 ppm gives 8 μs over 0.4 s and 200 μs over 10 s. These are examples, not measured errors of this lab's AWG. The 33220A's specified one-year internal reference accuracy is ± 20 ppm. [8]

If Option 001 is installed, share a 10 MHz reference. Its input accepts 100 mVpp–5 Vpp and requires lock acquisition before the run. [8] Presto can supply the reference; API 2.17 changed that output to disabled by default, so explicitly configure `hardware.set_output_reference(10e6)`. [4] [6]

Use one reference master. With Presto as master, keep its internal reference selection and feed its reference output to the AWG reference input. If another master already supplies the rig, distribute that clock consistently. Verify the installed option, signal level, and lock status. Reference locking controls rate drift; the proposed start event still establishes sweep phase.

Rounded folding periods cause a different drift

Current folding uses $N = \text{round}(f_s/f_r)$ samples per period. The timing error at the beginning of block m is:

$$\Delta t(m) = m * [N / f_s - 1 / f_r]$$

For $f_s = 50$ kHz and $f_r = 300$ Hz, $N = 167$. The block boundary slips by 6.667 μs per period; at block 199 it has slipped by 1.327 ms, about 40% of a ramp. A shared reference does not correct this arithmetic mismatch. Smearing depends on feature width; there is no universal contrast-loss percentage.

Choose one defensible analysis method

Commensurate acquisition: after tuning Presto, choose an achievable AWG rate close to f_s/N , verify the physical ratio, and check residual drift over the full record. Construct capture length using the tuned rate and the intended complete periods.

Measured-phase folding: derive ramp phase from a witness or calibrated physical frequency, preserve the initial phase and discarded samples, and interpolate or bin against phase. Reject incomplete cycles and reset transients. A SCPI frequency query verifies the programmed setting; it does not independently measure the oscillator's true frequency.

Alternatives if the small change fails

Method	Why consider it / what remains
Record actual ramp phase	Retains continuous Lockin. Recovers voltage labels despite asynchronous starts and measures drift. Requires an independent timing witness and calibrated acquisition path.
Pulsed sequence	Schedules a marker and acquisition in the same hardware sequence. Requires a measurement redesign and a budget for bandwidth, memory, and averaging.
External SYSREF	Vendor-supported synchronization hardware could release both armed operations. Requires confirmed hardware, routing, and arm/start semantics.
Presto DC waveform	Removes the separate AWG from the bias path. Its waveform-start relation to Lockin pixels is not established by the reviewed public documentation.

Most robust diagnostic: observe the ramp in the pixel record

An engineered witness can compare the real bias ramp against a known threshold, use that comparator to switch an RF carrier, and record the carrier envelope on a spare Presto input. A unipolar sawtooth gives a recurring reference crossing. Calibrate comparator and RF-switch delay, choose a resolvable envelope duration, and choose a threshold crossing late enough to remain visible after the initial-pixel discard. Capture gate prehistory on the scope. This is a proposed measurement circuit, not existing wrapper functionality.

The documented Presto RF inputs are AC-coupled RF paths; directly digitizing a 500 Hz DC ramp or a TTL line through them is not a valid general solution. An RF witness, or a suitable DC digitizer synchronized to a recorded acquisition reference, is needed. [2] (p.3).

Hardware-timed acquisition

Pulsed exposes output_digital_marker and store on the sequence time grid; long flat matching and downsampling also merit evaluation. [5] The older hardware sheet gives a 524 μs full-rate contiguous store window. [2] Treat that as a baseline: current downsampling changes bandwidth and duration. Check usable IF bandwidth, channels, raw storage, integration, and repetition limits for the actual QC experiment.

An illustrative redesign would arm a finite AWG burst, issue one sequenced marker, and acquire at a calibrated offset. Triggered bursts accept 1–50,000 cycles or infinity; triggers during an active burst are ignored. [3] (pp.110, 113). Ensure rearm margin and preserve individual cycles if stochastic parity dynamics matter; averaging them in hardware changes the measurement.

Bench validation that can falsify the proposal

No bench test was run for this report. The following protocol is intended to distinguish a real common start from a merely repeatable-looking software sequence. Begin on a suitable test load and preserve raw calibration data.

1. Characterize the physical signals

Observe the gate at the AWG connector and the analog ramp simultaneously. Record idle level, first-ramp phase, trigger-to-ramp delay, low glitches, actual period, and behavior when the gate falls. Test the present 30 ms overlap configuration at the intended df values. Establish adequate settling and rearm intervals.

2. Put timing evidence inside the acquisition

Use a calibrated witness visible in the same returned Lockin pixels as the readout (p.7). A scope showing gate and AWG output alone cannot locate the first returned pixel. A spare scope channel or another marker emitted at every demodulation window also does not automatically identify which window get_pixels returns.

3. Insert deliberate host delay

Compare the current sequence with the proposal. After apply_settings(auto_sync=False), insert randomized waits spanning 0–100 ms before get_pixels, including values that are not integer ramp periods. Verify that the gate and ramp remain inactive while waiting and that the first recorded ramp event keeps the same sample position after release. A changing position falsifies the intended common-start behavior.

4. Measure repeatability and drift

As a first validation set, take at least 100 starts across reconnects and representative host load; then exercise the longest planned record and several rates. Capture pre-start baseline on the scope and enough post-start time to expose missed cycles. The simultaneous Lockin start does not supply pretrigger pixels. Report mean offset, standard deviation, range, outliers, cycle slips, and phase slope within records. Use absolute first-event timing as well as phase modulo one period; a whole-cycle error can otherwise be invisible.

5. Set acceptance from the scientific requirement

$\text{allowed timing uncertainty} = \text{allowed voltage error} / (V_{pp} * f_r)$

The witness's own uncertainty must be below that allowance. One-pixel repeatability at 50 kHz establishes only a coarse 20 μs bound, not 1 μs accuracy. Fit sub-window timing only with a calibrated response model and uncertainty estimate, or validate at a suitable higher rate. Separately account for finite-window averaging and bias-line distortion.

Pass: no early ramp, stable calibrated pixel-to-ramp offset, no unexplained cycle slips, and acceptable full-record drift. **Fail:** retain the raw evidence; use measured-phase acquisition or a validated hardware-sequenced redesign instead of adding a fitted constant sleep.

Boundaries of a later implementation

Changes to consider only after the timing test

Scope a future synchronization option to QC ramp startup in TimeStream and its callers. Preserve immediate output muting and AWG output-off behavior on success and failure. Include preparation and deferred arming in cleanup coverage. Account for changed on_acquire semantics: it currently promises a callback after generation has begun, which would no longer be true in the deferred-start path.

Save the synchronization method, actual API and connected firmware versions, tuned df, requested/raw/retained pixel counts, D, measured t_c , calibration uncertainty, AWG settings and reference state, and phase/drift model. Keep untrimmed data. Save a voltage or phase coordinate with the folded trace so later analysis does not silently reinterpret its zero.

Two tempting routes that remain unproven

AWG Sync → Presto digital input: ordinary digital-input triggering is documented for Pulsed. [4] It is not interchangeable with Lockin's external SYSREF mechanism. Merely setting `get_pixels(auto_sync=False)` does not select a TTL input. A vendor-confirmed external synchronization design needs explicit clock, SYSREF, and readiness handling.

Built-in DC waveform: Presto 2.17 added repeating DC-bias waveforms, including a sawtooth. [6] [7] The reviewed interface does not establish the start phase relative to returned pixels. Before replacing the AWG, verify that timing, update cadence, reset transients, and analog bandwidth meet the requirement. Its DC output impedance is 1 k Ω , so loading also differs from the AWG path. [2]

Questions for Intermod, if bench evidence is ambiguous

Does the proposed `apply_settings(auto_sync=False) → get_pixels(auto_sync=True)` sequence release generation and pixel integration from the same SYSREF? Is digital output guaranteed low before that event? What is the first returned pixel's exact integration interval, including the skipped invalid pixel? Is trigger width greater than a window a supported continuous gate? How should trigger-only updates be applied in 2.17.1? Can a DC waveform reset its phase on that same event?

These are suggested technical questions; no message was sent to the manufacturer.

Conclusion

There is a technically grounded path forward. The deferred-generation change is the best first test and could remove the O(50 ms) variable start gap with limited wrapper changes. Reliable voltage assignment additionally requires a measured pixel origin, preserved trim phase, and control or measurement of drift. I cannot honestly claim that synchronization is fixed until the bench tests establish those properties.

Sources and audit trail

Official documentation accessed 15 September 2026. Numbered citations are clickable. Manufacturer facts support the design; proposed circuits, error calculations, and test criteria are engineering analysis. Page numbers for the 33220A guide are its printed numbers; the PDF viewer page number is one larger.

[1] Presto Lockin API, 2.17.1

<https://www.intermod.pro/manuals/presto/source/lockin.html>

apply_settings, get_pixels, set_trigger_out; separate auto_sync controls.

[2] Presto hardware datasheet, 2024-07-15

https://intermod.pro/res/docs/presto_spec_sheet.pdf

pp.2-5: pulse resources, analog inputs, digital ports, clocks and DC outputs.

[3] Agilent 33220A User's Guide, edition 4, May 2007

<https://www.keysight.com/us/en/assets/9018-04437/user-manuals/9018-04437.pdf>

pp.68-69, 107-114, 144-145, 223-230, 347-348: Sync, burst, phase, reference and trigger behavior.

[4] Presto Hardware API, 2.17.1

<https://www.intermod.pro/manuals/presto/source/hardware.html>

set_output_reference, TriggerSource, setup_dcbias_waveform.

[5] Presto Pulsed API, 2.17.1

<https://www.intermod.pro/manuals/presto/source/pulsed.html>

output_digital_marker, store, downsampling, flat matching and run.

[6] Presto changelog

<https://www.intermod.pro/manuals/presto/CHANGELOG.html>

2.12.0 auto_sync; 2.17.0 DC waveforms and reference-output defaults.

[7] Presto DC-bias waveforms, 2.17.1

<https://www.intermod.pro/manuals/presto/source/dcbias.html>

DcBiasWaveform interface and Lockin example; no explicit pixel-origin guarantee.

[8] 33220A product datasheet, 5988-8544

<https://www.keysight.com/us/en/assets/7018-01144/data-sheets/5988-8544.pdf>

Reference accuracy and Option 001 electrical specifications.

Local code inspected

Repository: C:/Users/AQD/Documents/GitHub/daq

Commit: 8f1e4d7700fb5e12fff5c3a9d299d88ab226465e (clean before report creation).

daq/measurements/timestream.py:106-124, 269-293, 296-419

daq/measurements/_gate_bias.py:162-200, 258-288

daq/measurements/qc_trace.py:240-272, 277-380

daq/measurements/sweep_std_dev.py:625-691

daq/instruments/function_generator.py:180-247; daq/analysis/folding.py:39-106

Installed source root: C:/Users/AQD/.conda/envs/presto/Lib/site-packages/presto

version.py:62-75; lockin.py:222-314, 548-598, 794-905; hardware.py:665-731.

The package identifies API 2.17.1 and expected conductor 0.11.5 / FPGA 48. These are not a query of the connected Presto. Repository CLAUDE.md was consulted for environment and hardware context; manufacturer documentation takes precedence over its older assumptions.